

touch-sensitive body cover enables the phone to understand the environment and adjust to it.

Morph, another Nokia concept, relies on nanotechnology to deliver flexible materials, transparent electronics and self-cleaning surfaces.

Fluid Smartphone, designed by Brazilian designer Dinard da Mata for Philips, works like a slap-wrap bracelet that wraps around the wrist.

Samsung has unveiled several ideas using its YOUNG flexible OLED displays. So far, LG too has shown off many wonderful flexible concepts like Flutter and Helix bracelet.

Apple, too, has filed patents for flexible devices. So we expect a lot of fun to unfurl in the coming years.

Wear your phone

A logical result of miniaturisation and flexibility are wearable phones—a theme that really excites concept phone designers. LG's Helix bracelet is a rubber and stainless-steel phone full of flexible electronics and a touchscreen display. When attached to a belt, the magnetic charger taps kinetic energy to charge the battery.

A wearable phone concept designed by Philipp Skorobogatov in 2013 turned out to be a finalist in the recent HTC 2020 design contest.

The wearable is basically a fancy bracelet with modular aspects, customisable looks and features. Future versions of this design will have energy-saving screens.

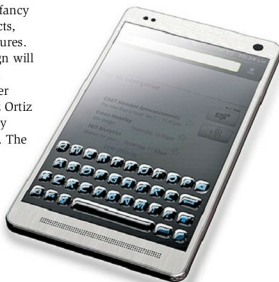
In 2010, Mexican designer Veronica Eugenia Rodriguez Ortiz created Yuxa, an eco-friendly bracelet-like phone concept. The device is made of materials like plant fibres and biodegradable plastic. It has an OLED display, rechargeable battery and an innovative way of communicating with people using vibrations.

Some designs like these continue to entice people years after their creation. However, others go in a pool. Glove One, for example, was a design developed by Bryan Cera in 2012. It was designed like a glove you wear, with components wrapping all your fingers. Too cumbersome!

In 2014, Big Bang Studios released a video showing a futuristic Ring Phone with a holographic projector, wireless charging battery and other nice ideas. However, comments on the video showed that users might not like the idea of a holographic display, which offers no privacy.

See-through phones

Last year, Lenovo-backed Chinese company ZUK showed off a concept phone that had a fully-transparent screen. The phone could perform all the functions of a regular smartphone and display all details on the screen, while still letting users see through it. The concept featured an almost bezel-free frame. However, it would take many more years to make a fully-functional phone of that sort, according to experts, because the battery itself



Buttons when you need these, gone when you do not (Image courtesy: Tactus Technology)

would take up a lot of space.

Fitting the battery and other hardware on a large bezel on the side of the transparent screen would look a little awkward in a small form device like a smartphone. That said, when we get to a stage when phones can be powered through Wi-Fi or other wireless means, then perhaps such a fully-transparent, bezel-free phone would become feasible.

Keyboard on demand

Although mobile devices are getting quite powerful, many people still find it difficult to type long documents using on-screen keyboards.

A technology being developed by Tactus Technology could provide a more comfortable way to type in future phones. The company's prototype touchscreen can show up a user-friendly QWERTY keyboard on demand. This involves replacing a touchscreen phone's window or cover lens with a special tactile layer, which works based on the principles of micro-fluidics.

According to the company, "Small fluid channels are routed throughout the tactile layer and



Phonebloks' modular phone (Image courtesy: Phonebloks)